

Simulation Hedy 2025

Team: TU Wien Space Team

Project: Lamarr

Rocket: Hedy

Launched on 12.10.2025 at EuRoC

Installs

In this section all needed libraries are installed and the needed classes imported

```
In [ ]: %pip install rocketpy
```

```
In [ ]: %pip install CoolProp
```

```
In [1]: from rocketpy import Environment, Rocket, Flight, Fluid, LiquidMotor, CylindricalTa
import CoolProp.CoolProp as CP
import numpy as np
```

Configuration

The length unit chosen here is millimeters to keep the values more readable. If necessary, values should be converted accordingly.

```
In [480... # Rocket configuration data
env_config = {
    "date" : (2025, 10, 12, 17, 00, 00), # yyyy, mm, dd, hh, mm, ss # preset
    "latitude" : 39.39046096801758, # Launch Latitude # preset
    "longitude" : -8.288534164428711, # Launch Longitude # preset
    "max_height" : 9000, # m # preset
    "timezone" : "Europe/Lisbon" # GMT+1 # preset
}
motor_config = {
    "holddown_time" : 1.4, # s # preset

    "nitrogen_tank" : {
        "length" : 235, # mm # measured
        "CG_lox" : 2653, # mm # measured # CG Lox nitrogen ta
        "CG_ethanol" : 1450, # mm # measured # CG ethanol nitroge
        "outer_diameter" : 93, # mm # measured
        "volume" : 1.2, # L # measured
        "massflow" : 0.03 # kg/s # estimated
    },
    "ethanol_tank" : {
        "length" : 573, # mm # measured
        "CG" : 852, # mm # measured
```

```

        "outer_diameter": 115,          # mm      # measured
        "volume" : 5.27,                # L        # measured
        "massflow" : 0.45                # kg/s     # tested
    },
    "lox_tank" : {
        "length" : 573,                  # mm      # measured
        "CG" : 1960,                     # mm      # measured
        "outer_diameter" : 115,          # mm      # measured
        "volume" : 5.27,                 # L        # measured
        "massflow" : 0.56                # kg/s     # tested
    },
    "nozzle" : {
        "diameter" : 75,                 # mm      # measured
        "position" : -40                  # mm      # measured
    },

    "thrust": "thrust_euroc_launch.csv", # N        # measured

    "temperature" : {
        "ethanol" : 298.15,              # K        # preset (= 25°C)
        "lox" : 93.15,                   # K        # preset (= -180°C)
        "nitrogen" : 298.15              # K        # preset (= 25°C)
    },
    "pressure" : {
        "ethanol" : 3000000,              # Pa       # preset
        "lox" : 3000000,                  # Pa       # preset
        "nitrogen" : 30000000             # Pa       # preset
    }
}

rocket_config = {
    "total_weight" : 17200,               # g        # weighed dry mass
    "total_length" : 3707,                # mm      # measured
    "total_CG" : 1813,                    # mm      # measured
    "moment_of_inertia_Z" : 0.0255,        # kg*m^2   # calculated
    "moment_of_inertia_XY" : 16.8,         # kg*m^2   # calculated
    "nosecone" : {
        "length" : 561,                   # mm      # measured    # 575 + 66 - 80 =
        "kind" : "lv haack"
    },
    "railbuttons" : {
        "upper" : 1646,                    # mm      # measured
        "lower" : 305                      # mm      # measured
    },
    "tailcone" : {
        "diameter" : 108,                  # mm      # measured
        "length" : 245,                    # mm      # measured
        "cylindrical_height" : 35          # mm      # measured
    },
    "rocket" : {
        "thickness" : 1.4,                 # mm      # measured
        "diameter" : 132.8                 # mm      # measured
    },
    "fins" : {
        "name" : "Biconvex-Freeform",
        "amount" : 4,

```

```

        "position" : 250,                # mm      # measured
        "shape_points" : ((0,0),
                           (0.250,-0.012),
                           (0.250, 0.108),
                           (0.195, 0.108),
                           (0,0))        # m      # measured
    },
    "parachutes" : {
        "main" : {
            "cd_s" : 6.911503837897546,    # calculated
            "trigger" : 450,                # m      # preset
            "sampling_rate" : 105,          # hz     # preset
            "lag" : 4,                      # s      # measured
            "noise" : (0, 8.3, 0.5)         # (pa, pa, pa) # preset
        },
        "drogue" : {
            "cd_s" : 0.5336875,
            "trigger" : "apogee",           # m      # preset
            "sampling_rate" : 105,          # hz     # preset
            "lag" : 1,                      # s      # measured
            "noise" : (0, 8.3, 0.5)         # (pa, pa, pa) # preset
        }
    }
}
flight_config = {
    "rail_length" : 11,                    # m      #Preset
    "inclination" : 84,                    # °      #Preset
    "heading" : 184,                       # °      #Preset
    "terminate_on_apogee" : False
}

```

Environments Initialization

In this section the environments are initialized.

- **envReanalysis**: environment with the weather data from the location and date of EuRoc'25

In [481...

```

#Ponte de Sor: 39.12368, -8.03333
#Launch date: 12.10.2025, 17:00

latitude   = env_config["latitude"]
longitude  = env_config["longitude"]
timezone   = env_config["timezone"]
date       = env_config["date"]
max_height = env_config["max_height"]

#Environment based on Forecast data for the EuRoC 2025
envReanalysis = Environment(max_expected_height = max_height)

envReanalysis.set_location(latitude = latitude, longitude = longitude)
envReanalysis.set_elevation("Open-Elevation")

```

```
envReanalysis.set_date(date, timezone = timezone)

envReanalysis.set_atmospheric_model(
    type="Reanalysis",
    file="euroc_weather.nc",
    dictionary="ECMWF"
)
envReanalysis.info()
```

Fetching elevation from open-elevation.com for lat=39.39046096801758, lon=-8.288534164428711...

Elevation received: 160.0 m

Gravity Details

Acceleration of gravity at surface level: 9.7798 m/s²

Acceleration of gravity at 6.555 km (ASL): 9.7601 m/s²

Launch Site Details

Launch Date: 2025-10-12 16:00:00 UTC | 2025-10-12 17:00:00 Europe/Lisbon

Launch Site Latitude: 39.39046°

Launch Site Longitude: -8.28853°

Reference Datum: SIRGAS2000

Launch Site UTM coordinates: 166021.44 W|E 0.00 N

Launch Site UTM zone: 31N

Launch Site Surface Elevation: 160.0 m

Atmospheric Model Details

Atmospheric Model Type: Reanalysis

Reanalysis Maximum Height: 6.555 km

Reanalysis Time Period: from 2025-10-12 15:00:00 to 2025-10-12 20:00:00 utc

Reanalysis Hour Interval: 1 hrs

Reanalysis Latitude Range: From 42.0° to 36.0°

Reanalysis Longitude Range: From -10.0° to -6.0°

Surface Atmospheric Conditions

Surface Wind Speed: 2.23 m/s

Surface Wind Direction: 67.67°

Surface Wind Heading: 247.67°

Surface Pressure: 996.64 hPa

Surface Temperature: 298.82 K

Surface Air Density: 1.162 kg/m³

Surface Speed of Sound: 346.54 m/s

Earth Model Details

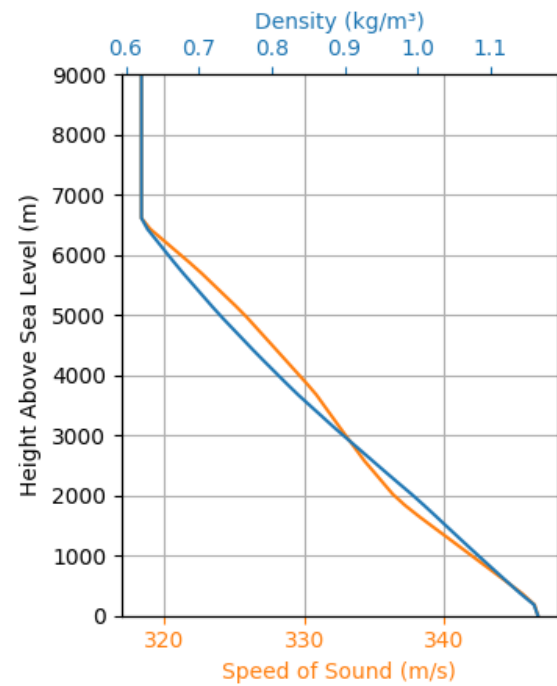
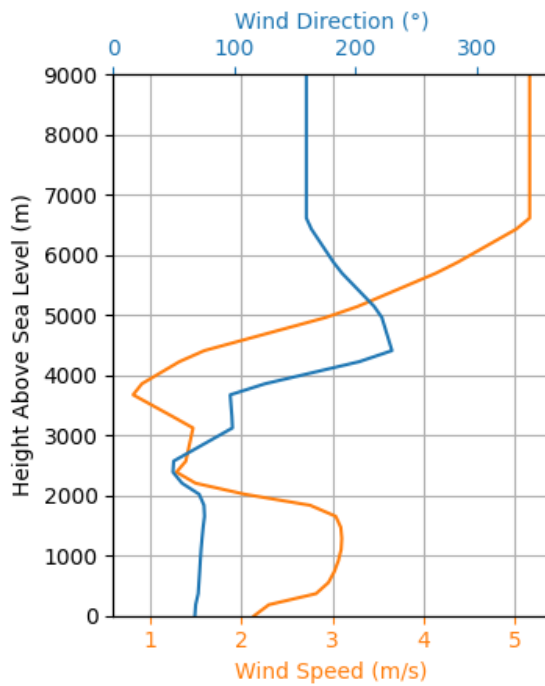
Earth Radius at Launch site: 6378.14 km

Semi-major Axis: 6378.14 km

Semi-minor Axis: 6356.75 km

Flattening: 0.0034

Atmospheric Model Plots



Simulation

Tanks / Engine

In [476...

```
#holddown time
t_holddown = motor_config["holddown_time"]

# tank height
h_nitrogen_tank = motor_config["nitrogen_tank"]["length"] / 1000
h_ethanol_tank = motor_config["ethanol_tank"]["length"] / 1000
h_lox_tank = motor_config["lox_tank"]["length"] / 1000
# OuterDiameter
OD_nitrogen_tank = motor_config["nitrogen_tank"]["outer_diameter"] / 1000
OD_ethanol_tank = motor_config["ethanol_tank"]["outer_diameter"] / 1000
OD_lox_tank = motor_config["lox_tank"]["outer_diameter"] / 1000
# volume
v_nitrogen_tank = motor_config["nitrogen_tank"]["volume"] / 1000
v_ethanol_tank = motor_config["ethanol_tank"]["volume"] / 1000
v_lox_tank = motor_config["lox_tank"]["volume"] / 1000
# massflows
mdot_nitrogen = motor_config["nitrogen_tank"]["massflow"]
mdot_ethanol = motor_config["ethanol_tank"]["massflow"]
mdot_lox = motor_config["lox_tank"]["massflow"]
# nozzle
nozzle_diameter = motor_config["nozzle"]["diameter"] / 1000

#thrust
thrust = motor_config["thrust"]

# Temperature Lox & Ethanol
T_nitrogen = motor_config["temperature"]["nitrogen"]
T_ethanol = motor_config["temperature"]["ethanol"]
```

```

T_lox                = motor_config["temperature"]["lox"]
# pressure LOX & Ethanol
p_nitrogen            = motor_config["pressure"]["nitrogen"]
p_ethanol             = motor_config["pressure"]["ethanol"]
p_lox                 = motor_config["pressure"]["lox"]

# Propellants

# define density
rho_nitrogen = CP.PropsSI("D", "T", T_nitrogen, "P|gas", p_nitrogen, "N2") #
rho_ethanol  = CP.PropsSI("D", "T|liquid", T_ethanol, "P", p_ethanol, "ethanol") #
rho_lox      = CP.PropsSI("D", "T|liquid", T_lox, "P", p_lox, "oxygen") #

# define fluids
nitrogen = Fluid(name = "N2", density = rho_nitrogen)
ethanol  = Fluid(name = "ethanol", density = rho_ethanol)
lox      = Fluid(name = "LOX", density = rho_lox)

# define tanks geometry
nitrogen_tank_shape = CylindricalTank(radius = OD_nitrogen_tank / 2, height = h_nit)
ethanol_tank_shape  = CylindricalTank(radius = OD_ethanol_tank / 2, height = h_ethan)
lox_tank_shape       = CylindricalTank(radius = OD_lox_tank / 2, height = h_lox_tank, sph

m_nitrogen = v_nitrogen_tank * rho_nitrogen # m
m_ethanol  = v_ethanol_tank * rho_ethanol   # m
m_lox      = v_lox_tank * rho_lox           # m

t_burn_nitrogen = (m_nitrogen / mdot_nitrogen)-0.001 # s

t_burn_ethanol = (m_ethanol / mdot_ethanol)
t_burn_lox     = (m_lox / mdot_lox)

if(t_burn_ethanol<t_burn_lox):
    t_burn = t_burn_ethanol
    print(f"using ethanol burn time ({t_burn_ethanol}). LOX burntime is {t_burn_lox}")
else:
    t_burn = t_burn_lox
    print(f"using lox burn time ({t_burn_lox}). Ethanol burntime is {t_burn_ethanol}")

#account for holddown
t_burn -= t_holddown

m_lox = mdot_lox * t_burn + 0.0001 # adding a little more so python do
m_ethanol = mdot_ethanol * t_burn + 0.0001
m_nitrogen = mdot_nitrogen * t_burn + 0.0001

# define tanks
nitrogen_tank = MassFlowRateBasedTank(
    name = "nitrogen tank",
    geometry = nitrogen_tank_shape,
    flux_time = t_burn,

```

```

        initial_liquid_mass = 0, # kg
        initial_gas_mass = m_nitrogen,
        liquid_mass_flow_rate_in = 0, # kg/s
        liquid_mass_flow_rate_out = 0, # kg/s
        gas_mass_flow_rate_in = 0, # kg/s
        gas_mass_flow_rate_out = lambda t: mdot_nitrogen,
        liquid = Fluid(name = "liquid", density = 0.0001), # ignore
        gas = nitrogen
    )
    ethanol_tank = MassFlowRateBasedTank(
        name = "fuel tank",
        geometry = ethanol_tank_shape,
        flux_time = t_burn,
        initial_liquid_mass = m_ethanol,
        initial_gas_mass = 0, # kg
        liquid_mass_flow_rate_in = 0, # kg/s
        liquid_mass_flow_rate_out = lambda t: mdot_ethanol,
        gas_mass_flow_rate_in = lambda t: mdot_nitrogen,
        gas_mass_flow_rate_out = 0, # kg/s
        liquid = ethanol,
        gas = nitrogen
    )

    lox_tank = MassFlowRateBasedTank(
        name = "oxidizer tank",
        geometry = lox_tank_shape,
        flux_time = t_burn,
        initial_liquid_mass = m_lox,
        initial_gas_mass = 0, # kg
        liquid_mass_flow_rate_in = 0, # kg/s
        liquid_mass_flow_rate_out = lambda t: mdot_lox,
        gas_mass_flow_rate_in = lambda t: mdot_nitrogen,
        gas_mass_flow_rate_out = 0, # kg/s
        liquid = lox,
        gas = nitrogen
    )

    skuld = LiquidMotor(
        dry_mass = 0, # kg
        dry_inertia = (0,0,0), # kg*m^2
        center_of_dry_mass_position = 0, # m

        nozzle_radius = nozzle_diameter / 2,
        nozzle_position = 0, # m

        thrust_source = thrust,
        burn_time = t_burn,
        coordinate_system_orientation = "nozzle_to_combustion_chamber"
    )

    skuld.add_tank(tank = ethanol_tank, position = motor_config["ethanol_tank"]["CG"]
    skuld.add_tank(tank = lox_tank, position = motor_config["lox_tank"]["CG"]
    skuld.add_tank(tank = nitrogen_tank, position = motor_config["nitrogen_tank"]["CG"]
    skuld.add_tank(tank = nitrogen_tank, position = motor_config["nitrogen_tank"]["CG"]

    skuld.all_info()
```


Warning: Adding spherical caps to the tank will not modify the total height of the tank 0.235 m. Its cylindrical portion height will be reduced to 0.142 m.

Warning: Adding spherical caps to the tank will not modify the total height of the tank 0.573 m. Its cylindrical portion height will be reduced to 0.4579999999999996 m.

Warning: Adding spherical caps to the tank will not modify the total height of the tank 0.573 m. Its cylindrical portion height will be reduced to 0.4579999999999996 m.

using ethanol burn time (9.225280880958138). LOX burntime is 10.662815399911576

Nozzle Details

Nozzle Radius: 0.0375 m

Motor Details

Total Burning Time: 7.825280880958138 s

Total Propellant Mass: 8.373 kg

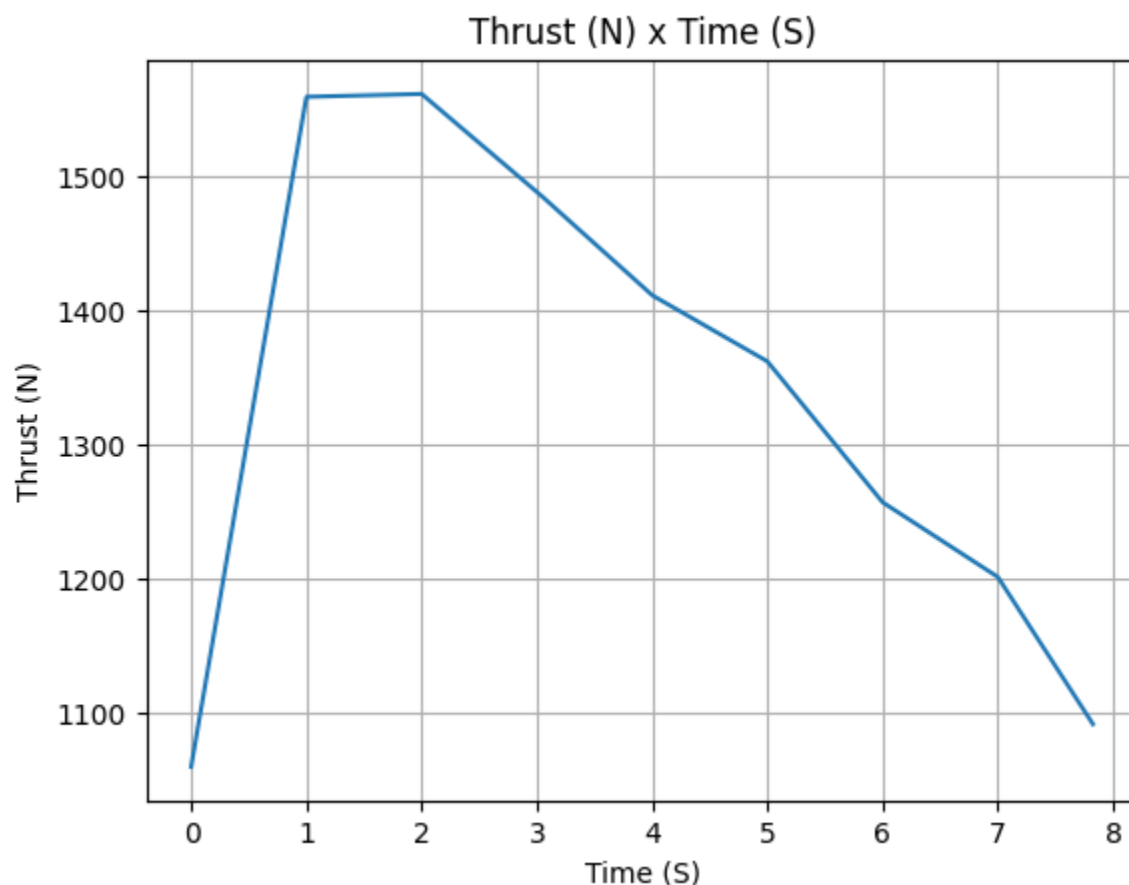
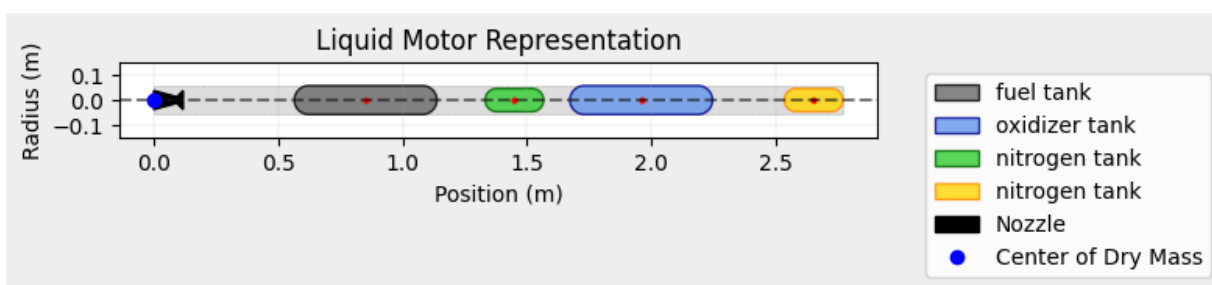
Structural Mass Ratio: 0.000

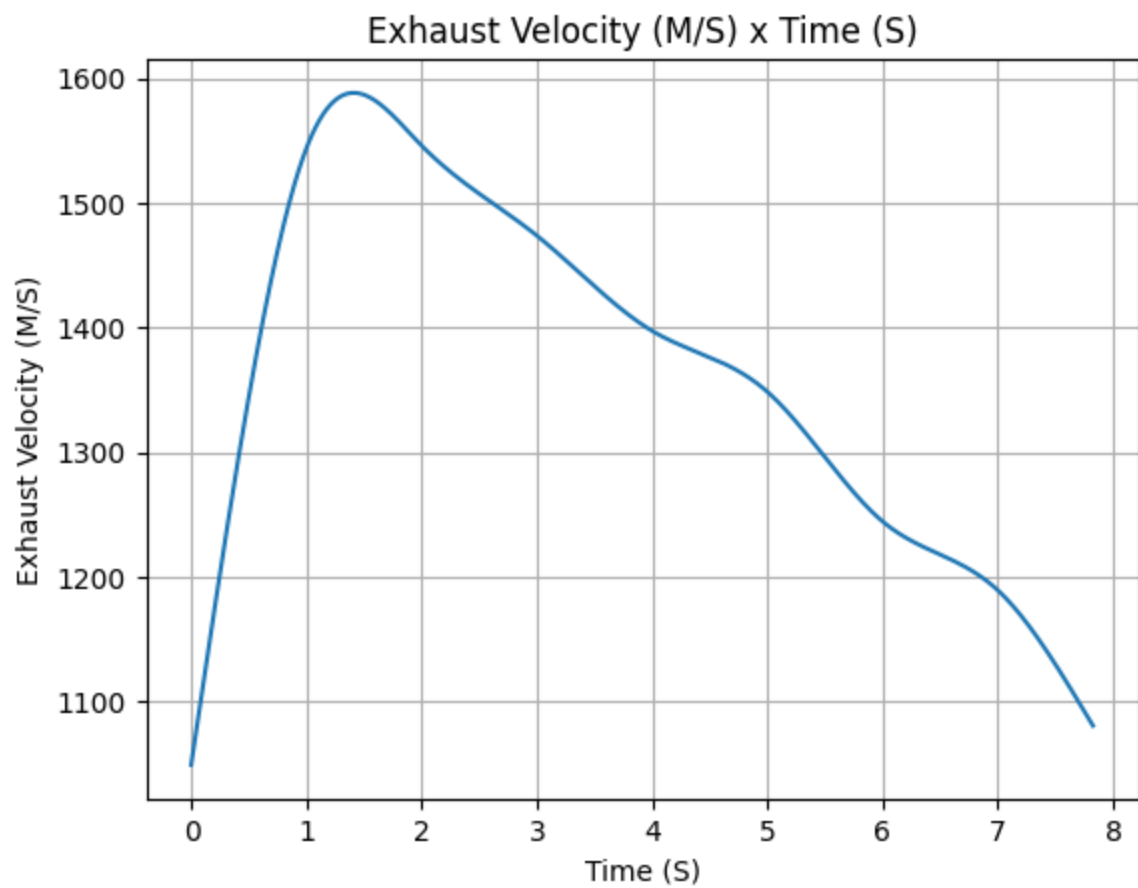
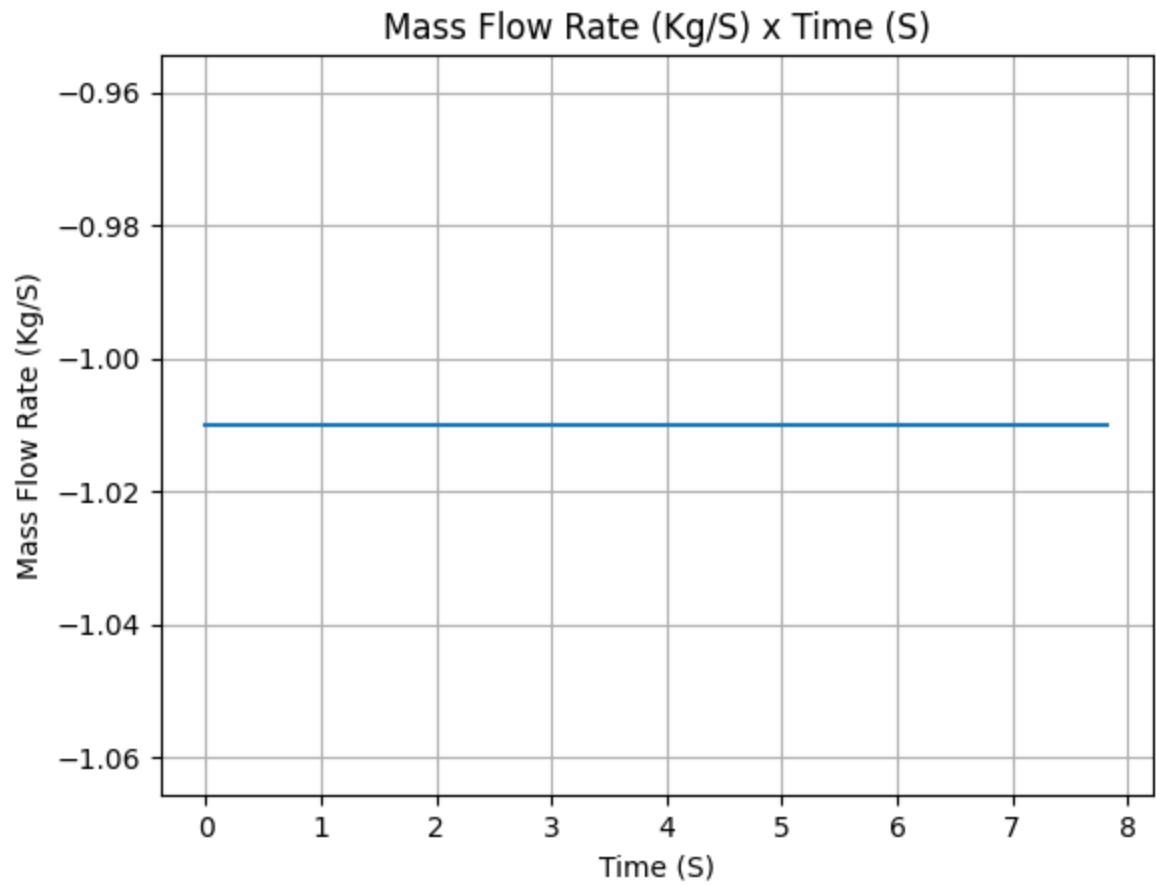
Average Propellant Exhaust Velocity: 1363.917 m/s

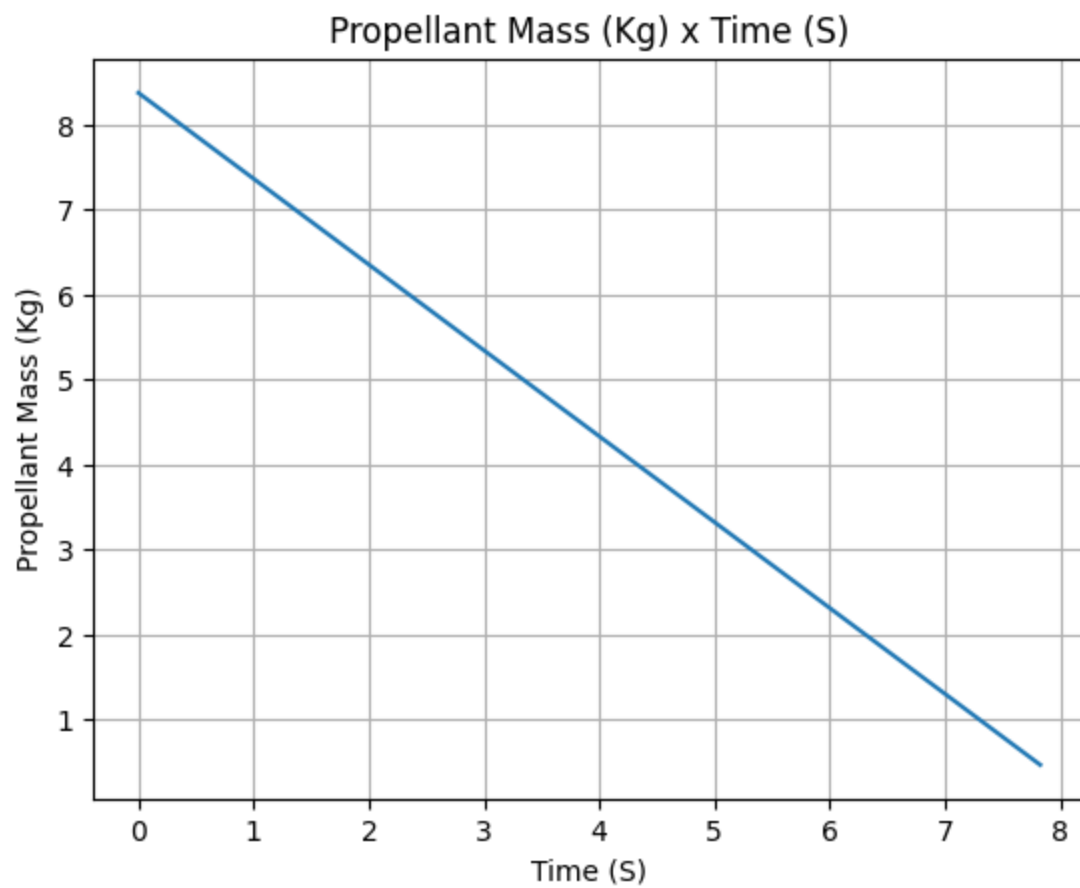
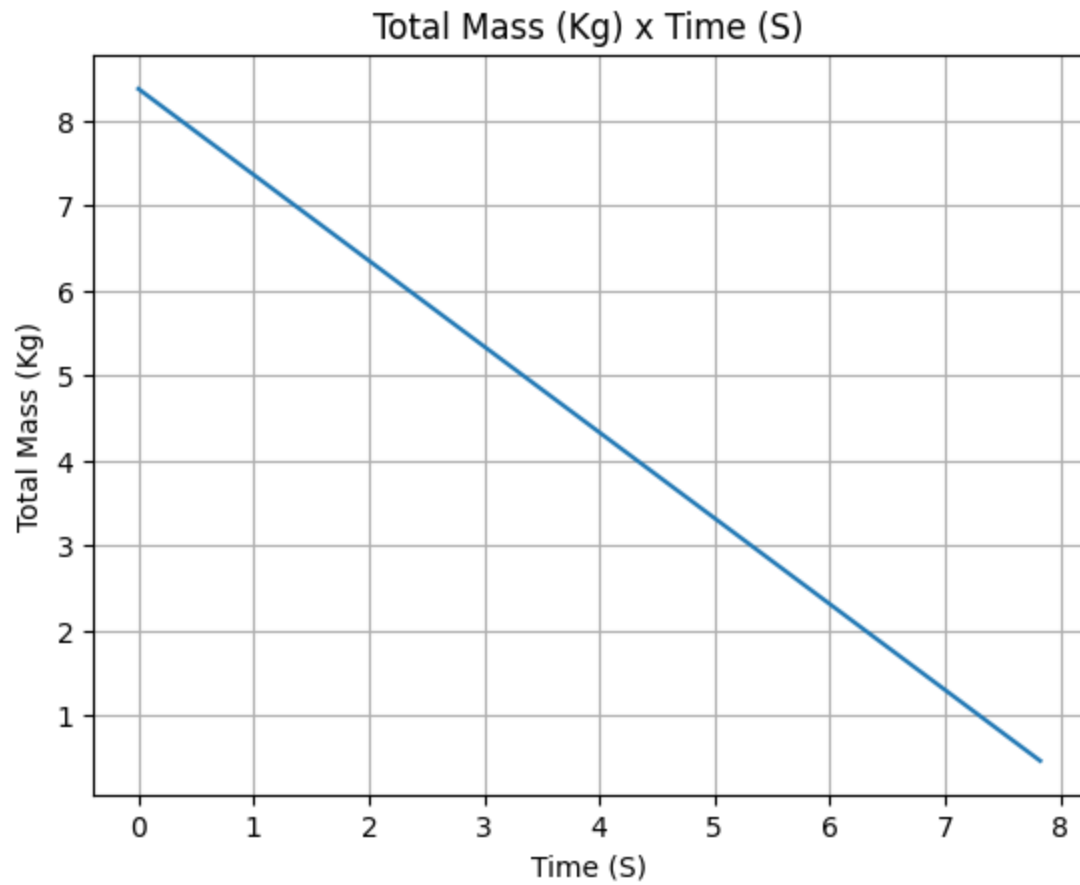
Average Thrust: 1369.450 N

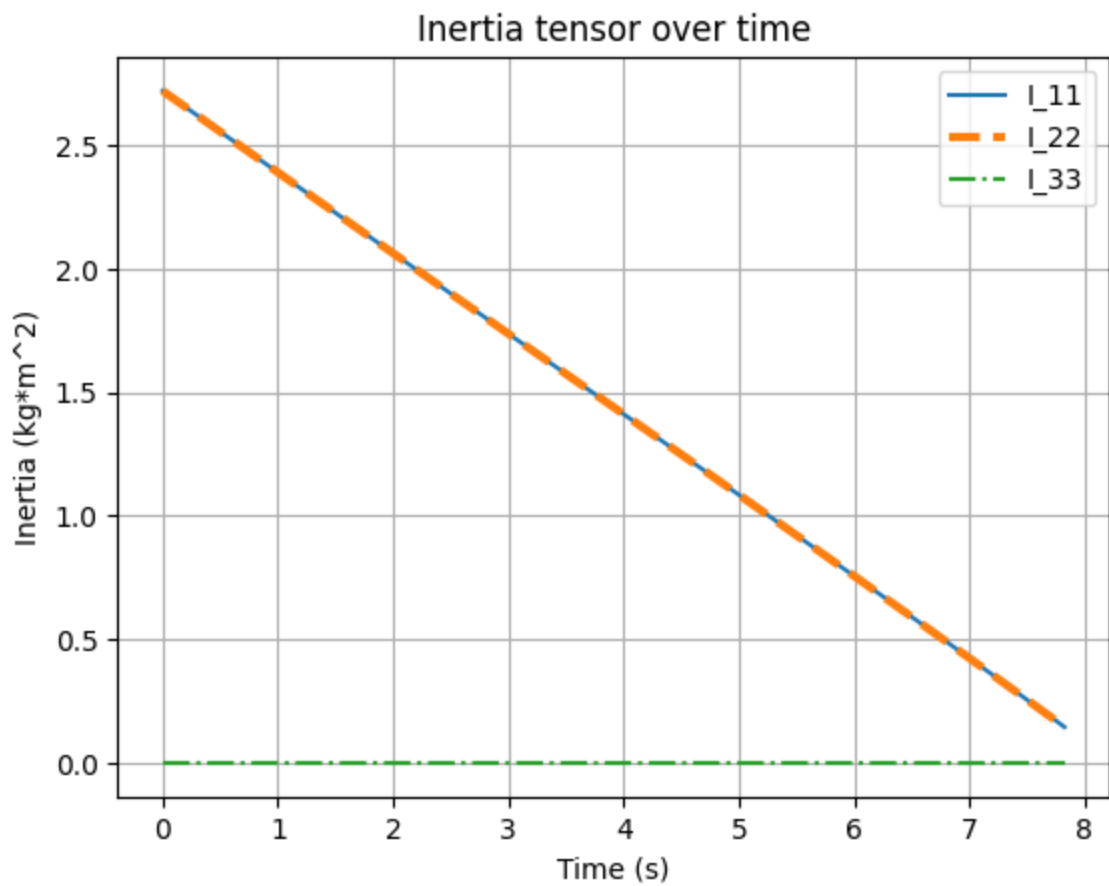
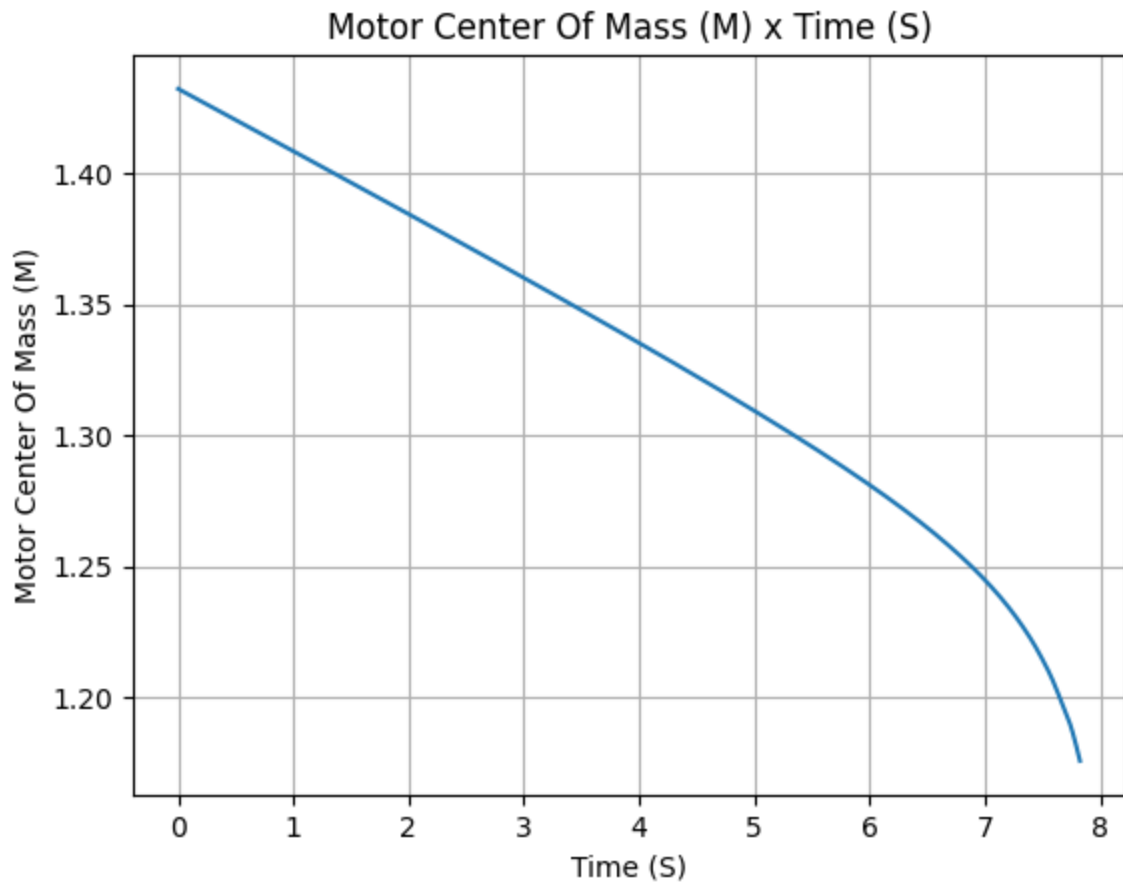
Maximum Thrust: 1561.34 N at 2.0 s after ignition.

Total Impulse: 10716.328 Ns









Rocket components

In [477...

```

#rocket
rocket_length      = rocket_config["total_length"]          / 1000
rocket_diameter    = rocket_config["rocket"]["diameter"]    / 1000
rocket_thickness   = rocket_config["rocket"]["thickness"]   / 1000

nosecone_length = (rocket_config["nosecone"]["length"]-80)  / 1000  # -80 mm to
nosecone_kind = rocket_config["nosecone"]["kind"]

nose_cone = NoseCone(
    length = nosecone_length,
    base_radius = rocket_diameter/2,
    kind = nosecone_kind
)

tailcone_cylindrical = rocket_config["tailcone"]["cylindrical_height"] / 1000
tailcone_length      = rocket_config["tailcone"]["length"]          / 1000
tailcone_bottom_radius = rocket_config["tailcone"]["diameter"]      /2 / 1000

tail = Tail(
    top_radius = rocket_diameter /2,
    bottom_radius = tailcone_bottom_radius,
    length = tailcone_length,
    rocket_radius = rocket_diameter/2
)

fin_amount      = rocket_config["fins"]["amount"]
fin_shape_points = rocket_config["fins"]["shape_points"]
fin_name        = rocket_config["fins"]["name"]

fin_set = FreeFormFins(
    n = fin_amount,
    shape_points = fin_shape_points,
    rocket_radius=tailcone_bottom_radius,
    name = fin_name
)
fin_set.draw()

main_cd_s      = rocket_config["parachutes"]["drogue"]["cd_s"]
main_trigger   = rocket_config["parachutes"]["main"]["trigger"]
main_sampling_rate = rocket_config["parachutes"]["main"]["sampling_rate"]
main_lag       = rocket_config["parachutes"]["main"]["lag"]
main_noise     = rocket_config["parachutes"]["main"]["noise"]

drogue_cd_s    = rocket_config["parachutes"]["drogue"]["cd_s"]
drogue_trigger = rocket_config["parachutes"]["drogue"]["trigger"]
drogue_sampling_rate = rocket_config["parachutes"]["drogue"]["sampling_rate"]
drogue_lag     = rocket_config["parachutes"]["drogue"]["lag"]
drogue_noise   = rocket_config["parachutes"]["drogue"]["noise"]

parachutes = {}

```

```

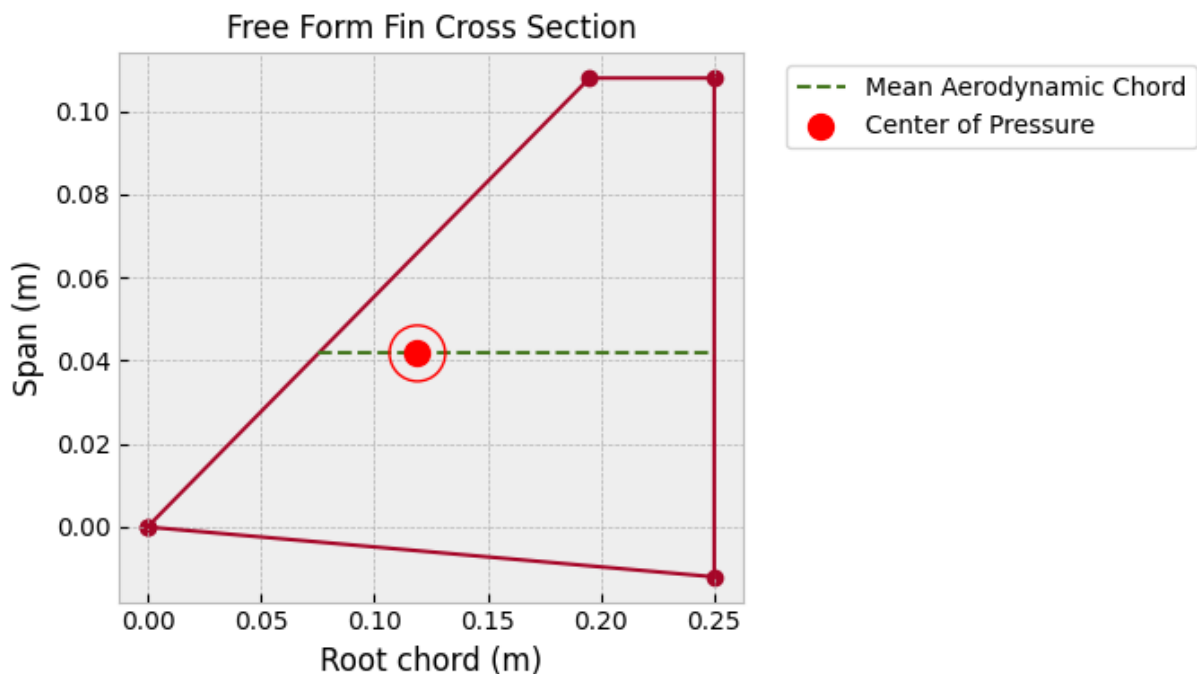
parachutes[0] = Parachute(
    name = "main",
    cd_s = main_cd_s,
    trigger = main_trigger,
    sampling_rate = main_sampling_rate,
    lag = main_lag,
    noise = main_noise
)

parachutes[1] = Parachute(
    name = "drogue",
    cd_s = drogue_cd_s,
    trigger = drogue_trigger,
    sampling_rate = drogue_sampling_rate,
    lag = drogue_lag,
    noise = drogue_noise
)

```

C:\Users\wattikan\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-packages\rocketpy\rocket\ aero_surface\ fins\ free_form_fins.py:143: UserWarning: Jagged fin shape detected. This may cause small inaccuracies center of pressure and pitch moment calculations.

warnings.warn(



Hedy

In [478...

```

total_mass          = rocket_config["total_weight"] / 1000

#inertia
inertia_x_y         = rocket_config["moment_of_intertia_XY"]
inertia_z           = rocket_config["moment_of_intertia_Z"]

upper_railbutton_position = rocket_config["railbuttons"]["upper"] / 1000
lower_railbutton_position = rocket_config["railbuttons"]["lower"] / 1000

```

```
fin_position          = rocket_config["fins"]["position"]      / 1000
nozzle_position        = motor_config["nozzle"]["position"]    / 1000

total_CG = rocket_config["total_CG"] / 1000

hedy = Rocket(
    radius = rocket_diameter / 2,
    mass = total_mass,
    inertia = (inertia_x_y, inertia_x_y, inertia_z),
    power_off_drag = "./power_off_drag.csv",
    power_on_drag = "./power_on_drag.csv",
    center_of_mass_without_motor = total_CG,
    coordinate_system_orientation = "tail_to_nose"
)

hedy.add_motor(skuld, position = nozzle_position)

hedy.set_rail_buttons(upper_button_position= upper_railbutton_position, lower_butto

hedy.add_surfaces(surfaces=[nose_cone, fin_set, tail], positions=[rocket_length, fi

hedy.parachutes = list(parachutes.values())

hedy.all_info()
```

Inertia Details

Rocket Mass: 17.200 kg (without motor)
Rocket Dry Mass: 17.200 kg (with unloaded motor)
Rocket Loaded Mass: 25.573 kg
Rocket Structural Mass Ratio: 0.673
Rocket Inertia (with unloaded motor) 11: 16.800 kg*m2
Rocket Inertia (with unloaded motor) 22: 16.800 kg*m2
Rocket Inertia (with unloaded motor) 33: 0.025 kg*m2
Rocket Inertia (with unloaded motor) 12: 0.000 kg*m2
Rocket Inertia (with unloaded motor) 13: 0.000 kg*m2
Rocket Inertia (with unloaded motor) 23: 0.000 kg*m2

Geometrical Parameters

Rocket Maximum Radius: 0.0664 m
Rocket Frontal Area: 0.013851 m2

Rocket Distances

Rocket Center of Dry Mass - Center of Mass without Motor: 0.000 m
Rocket Center of Dry Mass - Nozzle Exit: 1.853 m
Rocket Center of Dry Mass - Center of Propellant Mass: 0.421 m
Rocket Center of Mass - Rocket Loaded Center of Mass: 0.138 m

Aerodynamics Lift Coefficient Derivatives

Nose Cone Lift Coefficient Derivative: 2.000/rad
Biconvex-Freeform Lift Coefficient Derivative: 6.930/rad
Tail Lift Coefficient Derivative: -0.677/rad

Center of Pressure

Nose Cone Center of Pressure position: 3.436 m
Biconvex-Freeform Center of Pressure position: 0.131 m
Tail Center of Pressure position: 0.127 m

Stability

Center of Mass position (time=0): 1.675 m
Center of Pressure position (time=0): 0.932 m
Initial Static Margin (mach=0, time=0): 5.594 c
Final Static Margin (mach=0, time=burn_out): 6.496 c
Rocket Center of Mass (time=0) - Center of Pressure (mach=0): 0.743 m

Parachute Details

Parachute Main with a cd_s of 0.5337 m2
Ejection signal trigger: 450 m (AGL)
Ejection system refresh rate: 105.000 Hz
Time between ejection signal is triggered and the parachute is fully opened: 4.0 s

Parachute Details

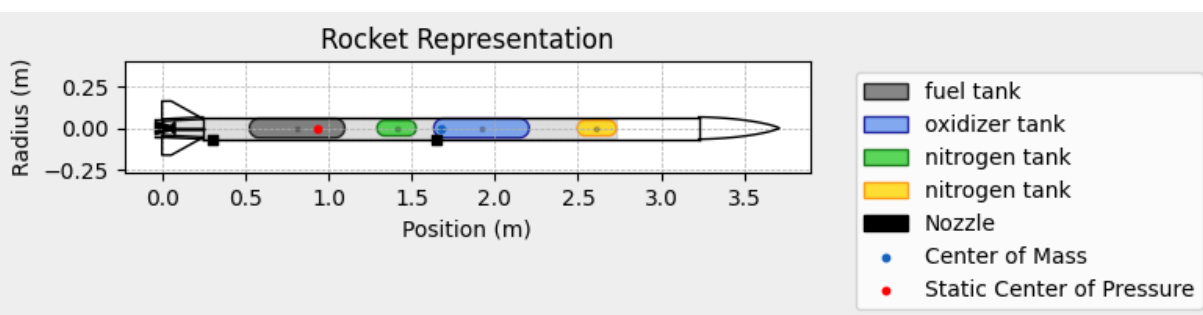
Parachute Drogue with a c_{d_s} of 0.5337 m²

Ejection signal trigger: At Apogee

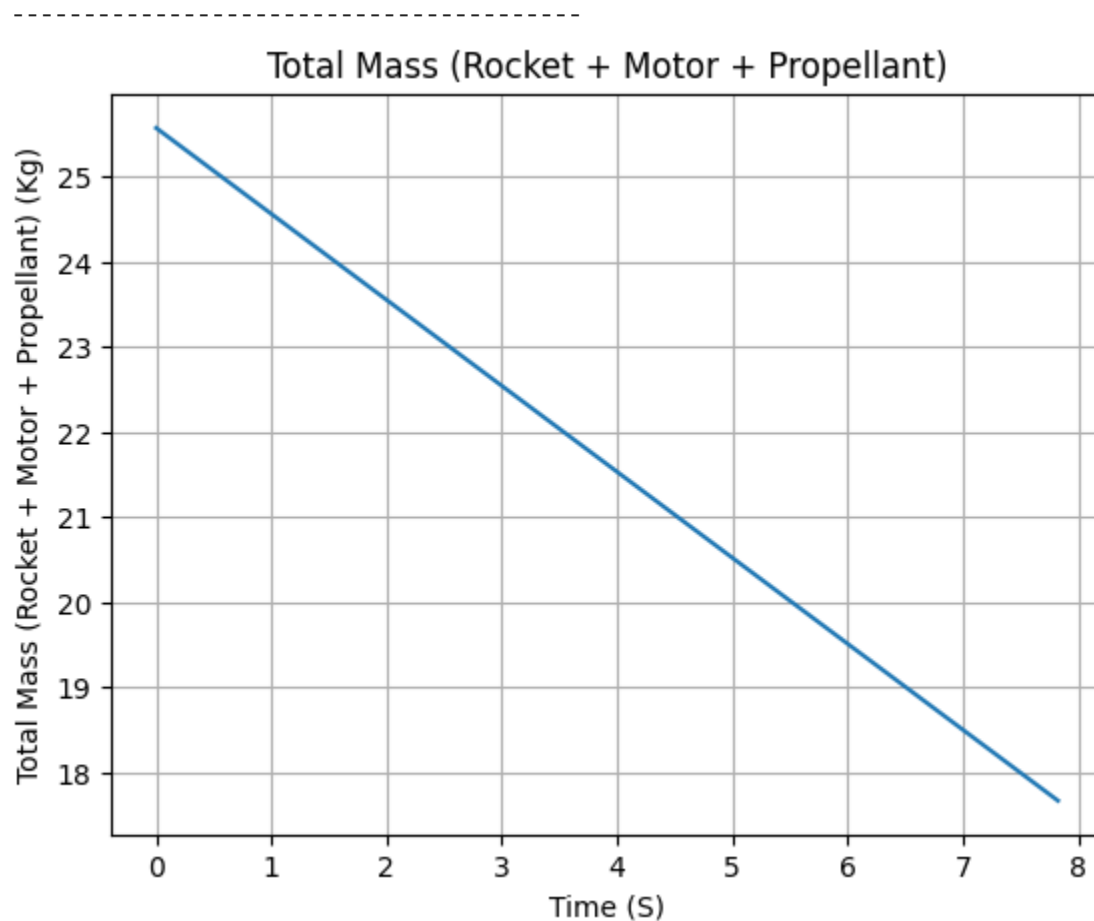
Ejection system refresh rate: 105.000 Hz

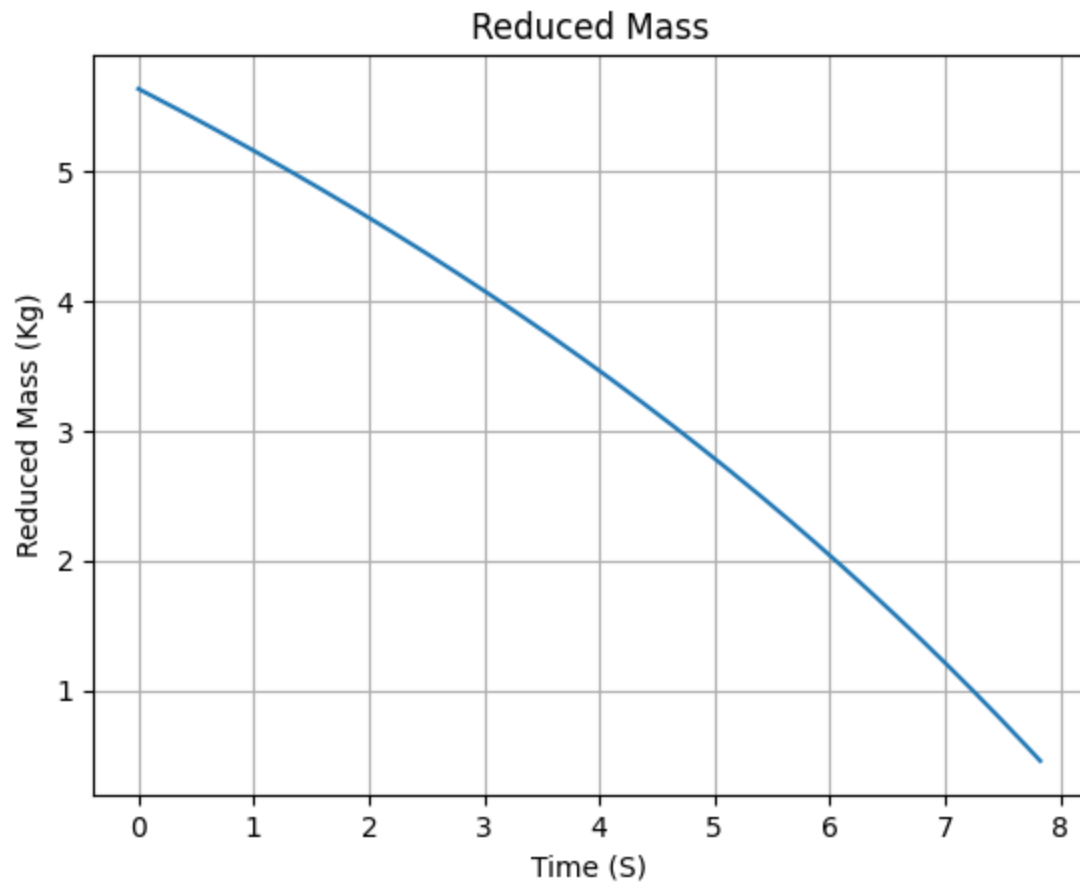
Time between ejection signal is triggered and the parachute is fully opened: 1.0 s

Rocket Draw



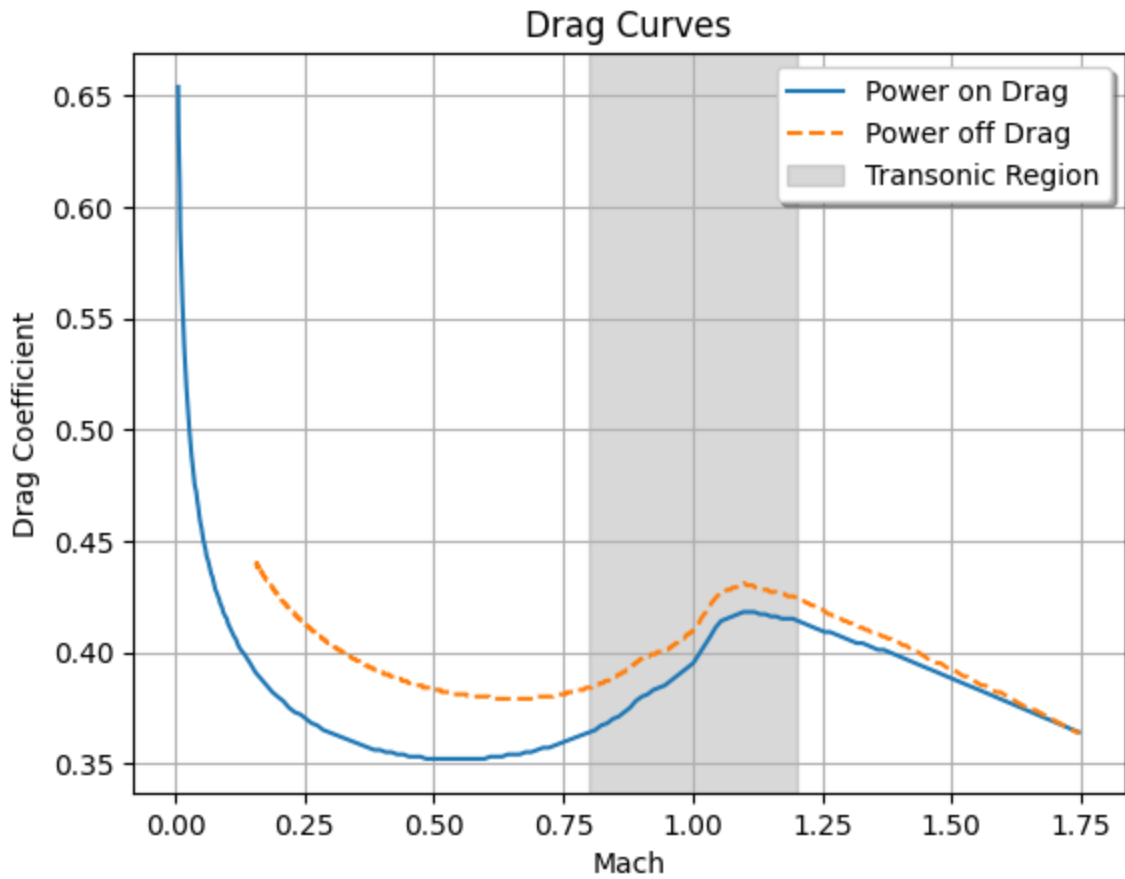
Mass Plots



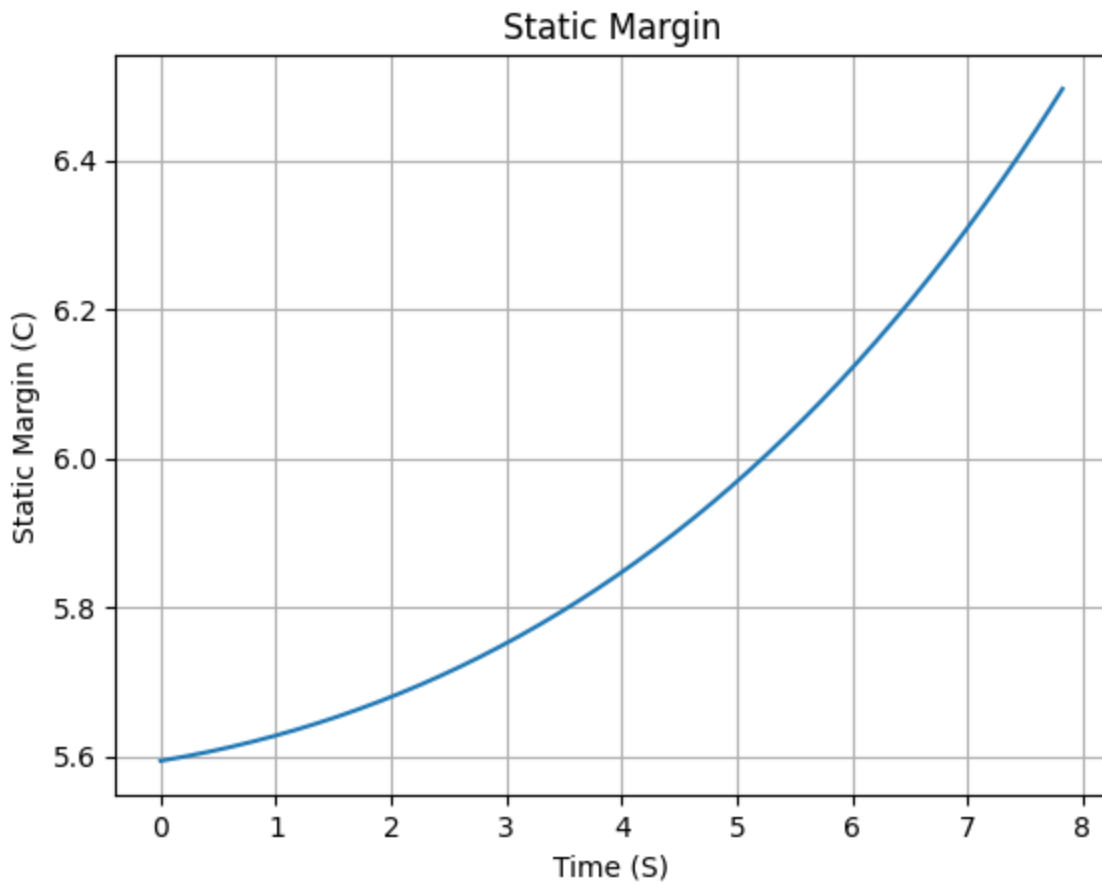


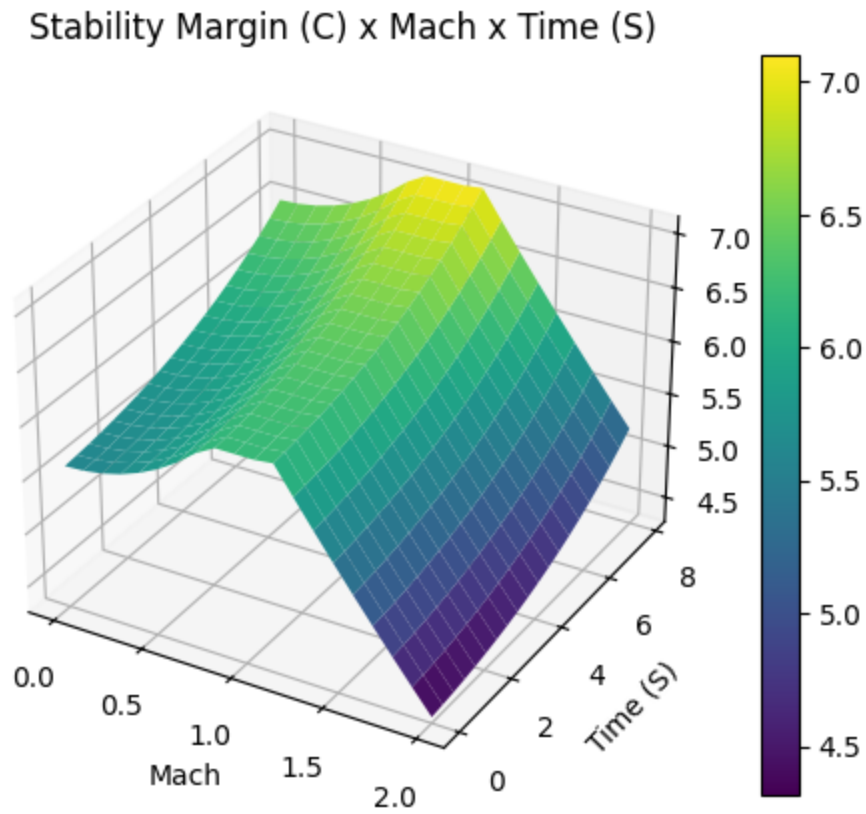
Aerodynamics Plots

Drag Plots

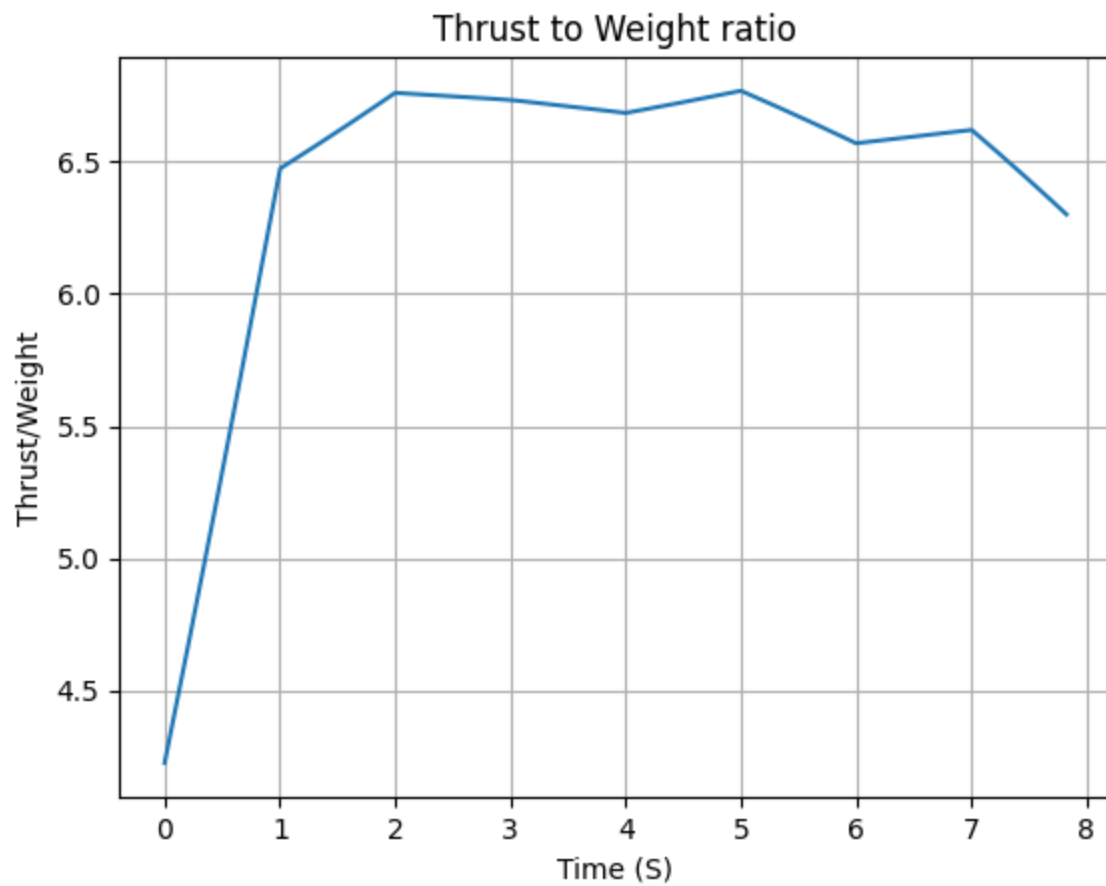


Stability Plots





Thrust-to-Weight Plot



Flight

In [479...

```
rail_length      = flight_config["rail_length"]
inclination      = flight_config["inclination"]
heading          = flight_config["heading"]
terminate_on_apogee = flight_config["terminate_on_apogee"]

flight_reanalysis = Flight(
    rocket        = hedy,
    environment    = envReanalysis,
    rail_length    = rail_length,
    inclination    = inclination,
    heading        = heading,
    terminate_on_apogee = terminate_on_apogee,
    name          = "Reanalysis"
)
print(f"Takeoff Mass: {hedy.total_mass(flight_reanalysis.out_of_rail_time)}")
flight_reanalysis.prints.out_of_rail_conditions()
flight_reanalysis.prints.apogee_conditions()
#flight_reanalysis.prints.impact_conditions()
flight_reanalysis.prints.maximum_values()
flight_reanalysis.plots.trajectory_3d()
flight_reanalysis.plots.stability_and_control_data()

#flight_reanalysis.prints.all()
#flight_reanalysis.plots.all()

#flight_reanalysis.all_info()

flight_reanalysis.export_kml()
```

Takeoff Mass: 24.854359706085653

Rail Departure State

Rail Departure Time: 0.712 s
Rail Departure Velocity: 28.027 m/s
Rail Departure Stability Margin: 5.622 c
Rail Departure Angle of Attack: 4.616°
Rail Departure Thrust-Weight Ratio: 5.807
Rail Departure Reynolds Number: 2.347e+05

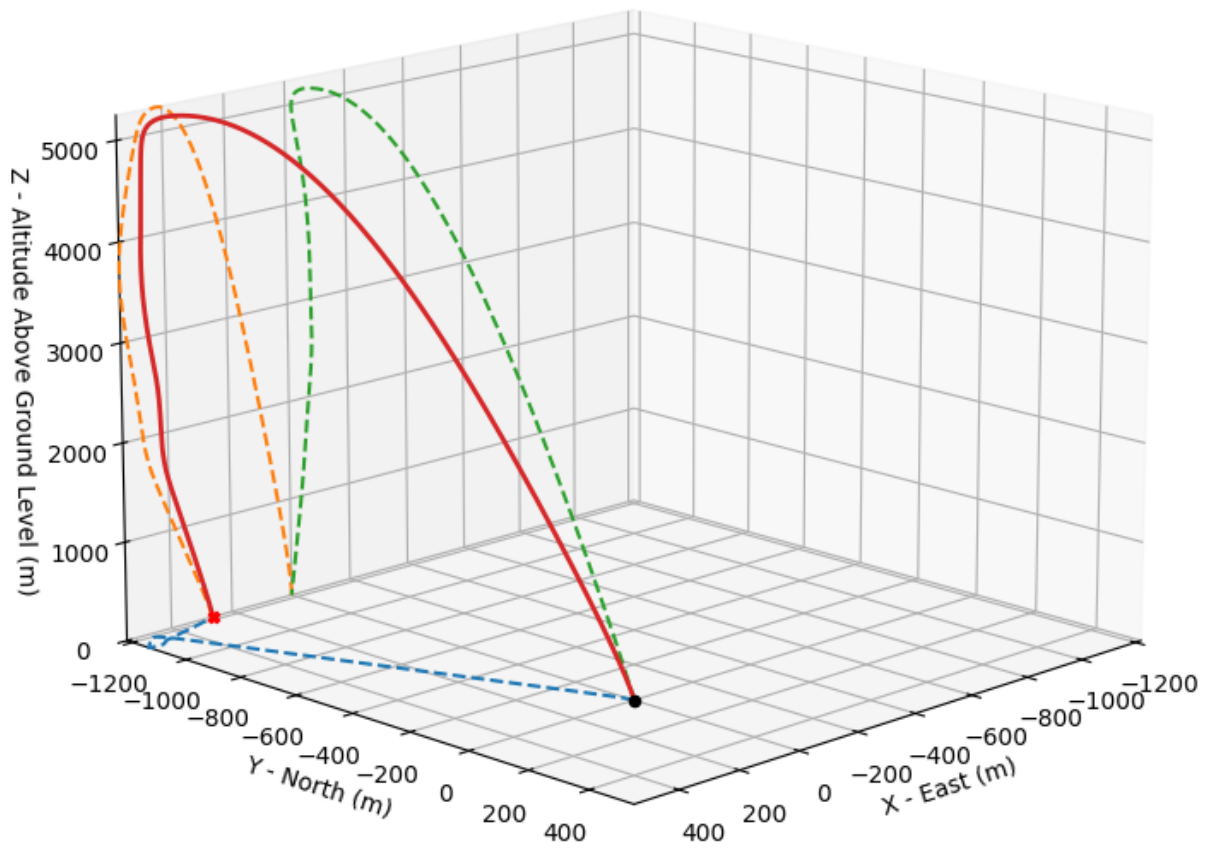
Apogee State

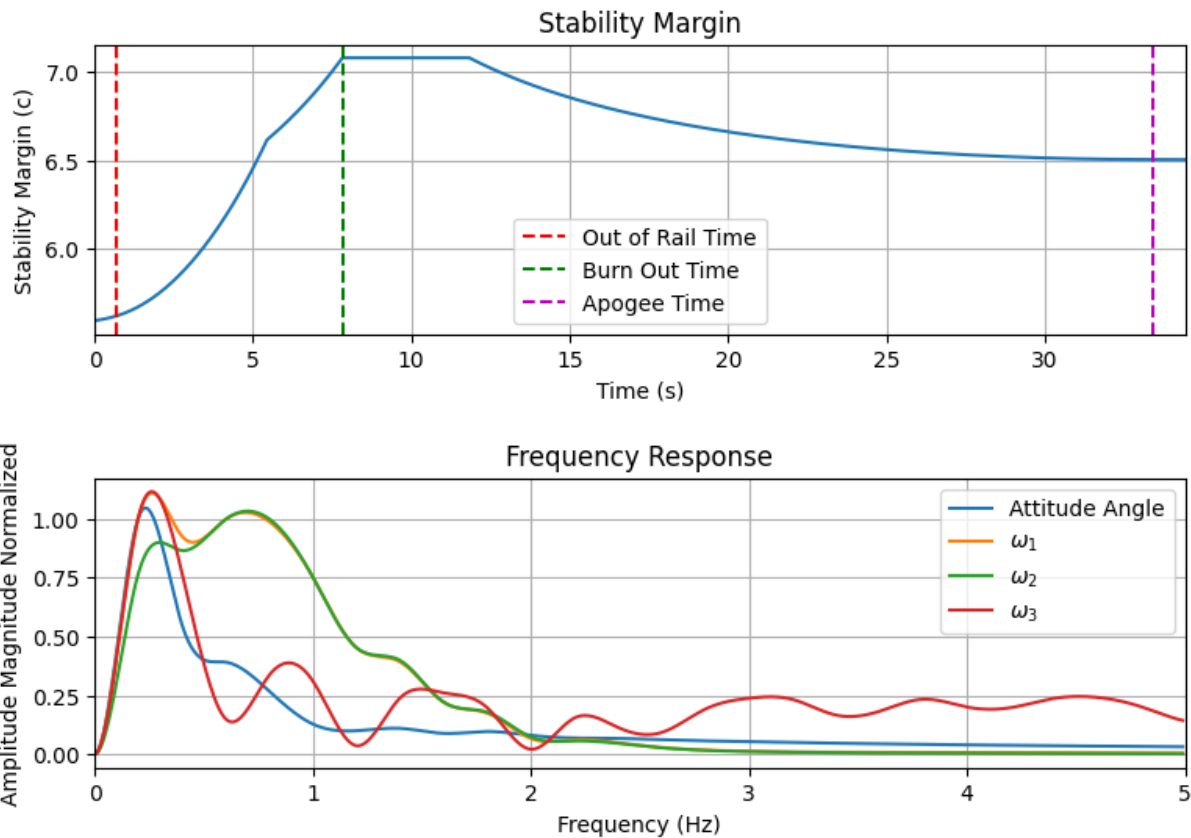
Apogee Time: 33.412 s
Apogee Altitude: 5406.908 m (ASL) | 5246.908 m (AGL)
Apogee Freestream Speed: 33.731 m/s
Apogee X position: 419.630 m
Apogee Y position: -1080.040 m
Apogee latitude: 39.3807587°
Apogee longitude: -8.2836572°

Maximum Values

Maximum Speed: 365.662 m/s at 7.83 s
Maximum Mach Number: 1.081 Mach at 7.83 s
Maximum Reynolds Number: 2.810e+06 at 7.83 s
Maximum Dynamic Pressure: 6.857e+04 Pa at 7.83 s
Maximum Acceleration During Motor Burn: 55.422 m/s² at 2.00 s
Maximum Gs During Motor Burn: 5.651 g at 2.00 s
Maximum Acceleration After Motor Burn: 31.722 m/s² at 0.00 s
Maximum Gs After Motor Burn: 3.235 Gs at 0.00 s
Maximum Stability Margin: 7.080 c at 7.83 s
Maximum Upper Rail Button Normal Force: 0.689 N
Maximum Upper Rail Button Shear Force: 1.417 N
Maximum Lower Rail Button Normal Force: 1.149 N
Maximum Lower Rail Button Shear Force: 2.353 N

Flight Trajectory





File trajectory.kml saved with success!

C:\Users\wattikan\AppData\Local\Temp\ipykernel_8884\850779798.py:30: DeprecationWarning: Moved to FlightDataExporter.export_kml() and will be removed in v1.12.0. Use `rocketpy.simulation.flight\_data\_exporter.FlightDataExporter.export\_kml` instead.
flight_reanalysis.export_kml()

Comparsion

Initialization

In [500...

```
from rocketpy.simulation.flight_data_importer import FlightDataImporter
from rocketpy import Function

def pressure_to_altitude(data):
    return 0.3048 * ((1 - (data / 1013.25) ** 0.190284) * 145366.45) # reference: h

columns_map_cats_vega = {
    "ts": "time",
    "filteredAltitudeAGL": "altitude",
    "filteredAcceleration": "az",
    "latitude": "latitude",
    "longitude": "longitude",
    "Ax": "acceleration_x",
    "Ay": "acceleration_y",
    "Az": "acceleration_z",
    "Gx": "gyro_x",
```



```
        "Gy": "gyro_y",
        "Gz": "gyro_z",
        "velocity": "speed",
        "P": "pressure"
    }

    cats_vega_flight = FlightDataImporter(
        name="CATS Vega Flight Data",
        paths=["CATS FLIGHT DATA/filteredDataInfo.csv", "CATS FLIGHT DATA/gnssInfo.csv"],
        columns_map=columns_map_cats_vega,
        units=None,
        interpolation="linear",
        extrapolation="zero",
        delimiter=",",
        encoding="utf-8",
    )

    columns_map_altimax = {
        "ZEIT": "time",
        "PRESS_FILTER": "pressure",
        "HEIGHT_FILTER": "altitude",
        "ACCEL": "acceleration",
        "SPEED": "speed"
    }

    altimax_flight = FlightDataImporter(
        name="Altimax Flight Data",
        paths=["ALTIMAX FLIGHT DATA/Hedy_Flug_EuRoC.csv"],
        columns_map=columns_map_altimax,
        units=None,
        interpolation="linear",
        extrapolation="zero",
        delimiter=",",
        encoding="utf-8",
    )

    columns_map_rcu = {
        "time": "time",
        "lora:gps_altitude:sensor": "altitude",
        "lora:gps_latitude:sensor": "latitude",
        "lora:gps_longitude:sensor": "longitude",
        "lora:gps_status:sensor": "gps_status",
        "lora:rcu_accel_x:sensor": "accel_x",
        "lora:rcu_accel_y:sensor": "accel_y",
        "lora:rcu_accel_z:sensor": "accel_z",
        "lora:rcu_gyro_x:sensor": "gyro_x",
        "lora:rcu_gyro_y:sensor": "gyro_y",
        "lora:rcu_gyro_z:sensor": "gyro_z",
        "lora:rcu_barometer:sensor": "pressure"
    }

    srad_flight = FlightDataImporter(
        name="SRAD Flight Data",
```

```

paths=["RCU FLIGHT DATA/2025-10-12_EuRoC_launch_filtered.csv"],
columns_map=columns_map_rcu,
units=None,
interpolation="linear",
extrapolation="constant",
delimiter=";",
encoding="utf-8",
)

```

The following attributes were create and are now available to be used: ['time', 'altitude', 'az']

The following attributes were create and are now available to be used: ['time', 'latitude', 'longitude']

The following attributes were create and are now available to be used: ['time', 'speed']

The following attributes were create and are now available to be used: ['time', 'acceleration_x', 'acceleration_y', 'acceleration_z', 'gyro_x', 'gyro_y', 'gyro_z']

The following attributes were create and are now available to be used: ['time', 'pressure']

The following attributes were create and are now available to be used: ['time', 'pressure', 'altitude', 'acceleration', 'speed']

The following attributes were create and are now available to be used: ['time', 'altitude', 'latitude', 'longitude', 'gps_status', 'accel_x', 'accel_y', 'accel_z', 'gyro_x', 'gyro_y', 'gyro_z', 'pressure']

Altitude

In [501...

```

data = srad_flight.pressure
altitude_srad = Function(data[np.isfinite(data[:,0]) & np.isfinite(data[:,1])])

apogee_srad = pressure_to_altitude(altitude_srad.min)

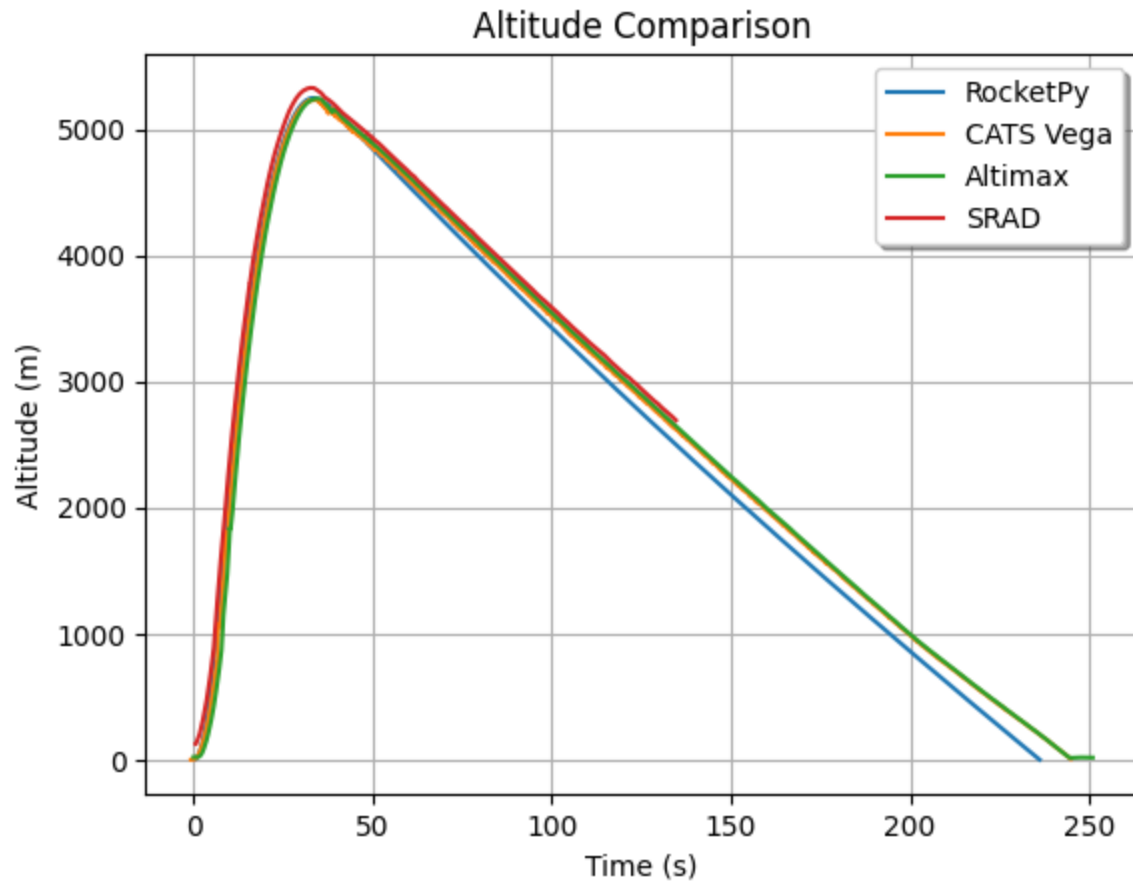
apogee_actual = (apogee_srad + cats_vega_flight.altitude.max) / 2
apogee_simulated = flight_reanalysis.apogee
apogee_error = abs(apogee_actual - apogee_simulated)
apogee_percentage_error = apogee_error / apogee_actual * 100

print(f"Actual apogee: {apogee_actual:.2f} m")
print(f"Simulated apogee: {apogee_simulated:.2f} m")
print(f"Error: {apogee_error:.2f} m")
print(f"Percentage Error: {apogee_percentage_error:.2f}%")

Function.compare_plots(
    [
        (flight_reanalysis.altitude, "RocketPy"),
        (cats_vega_flight.altitude, "CATS Vega"),
        (altimax_flight.altitude, "Altimax"),
        (pressure_to_altitude(altitude_srad), "SRAD")
    ],
    title="Altitude Comparison",
    xlabel="Time (s)",
    ylabel="Altitude (m)",
)

```

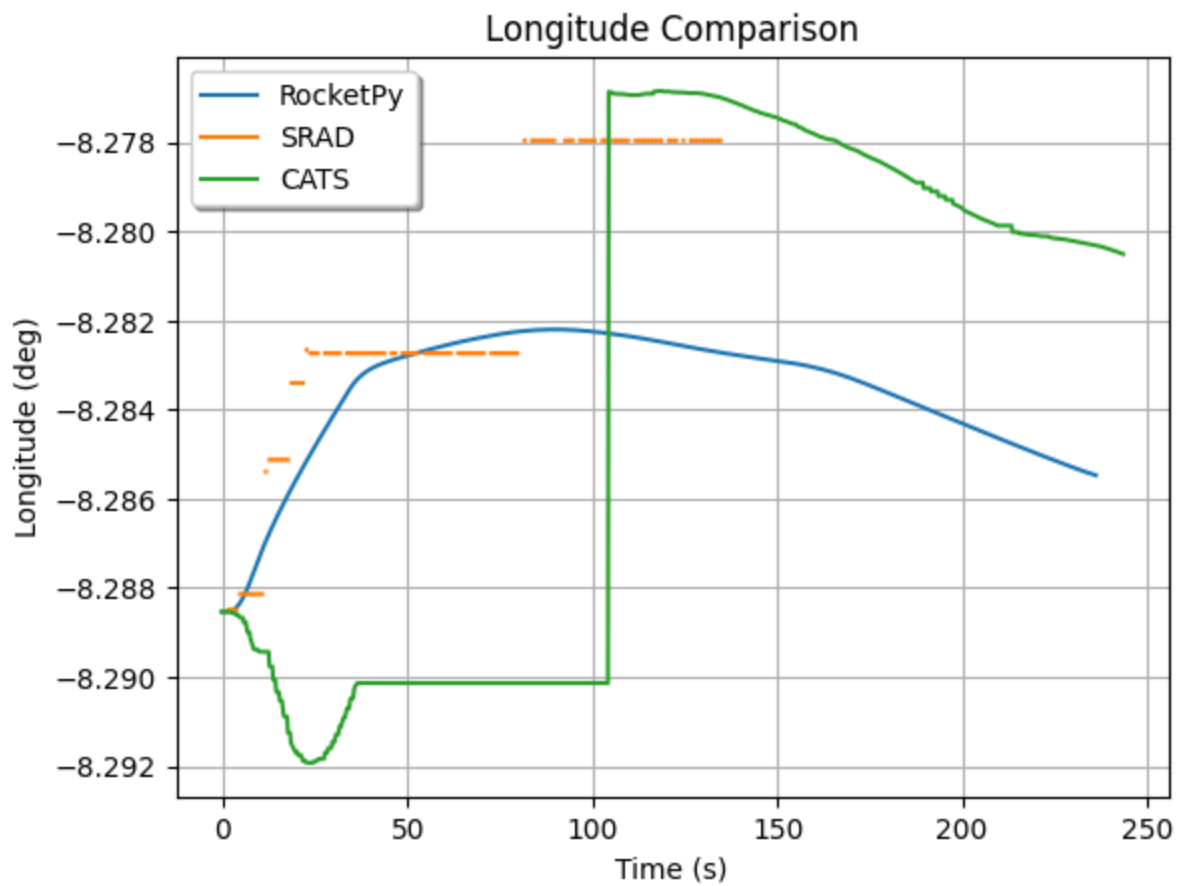
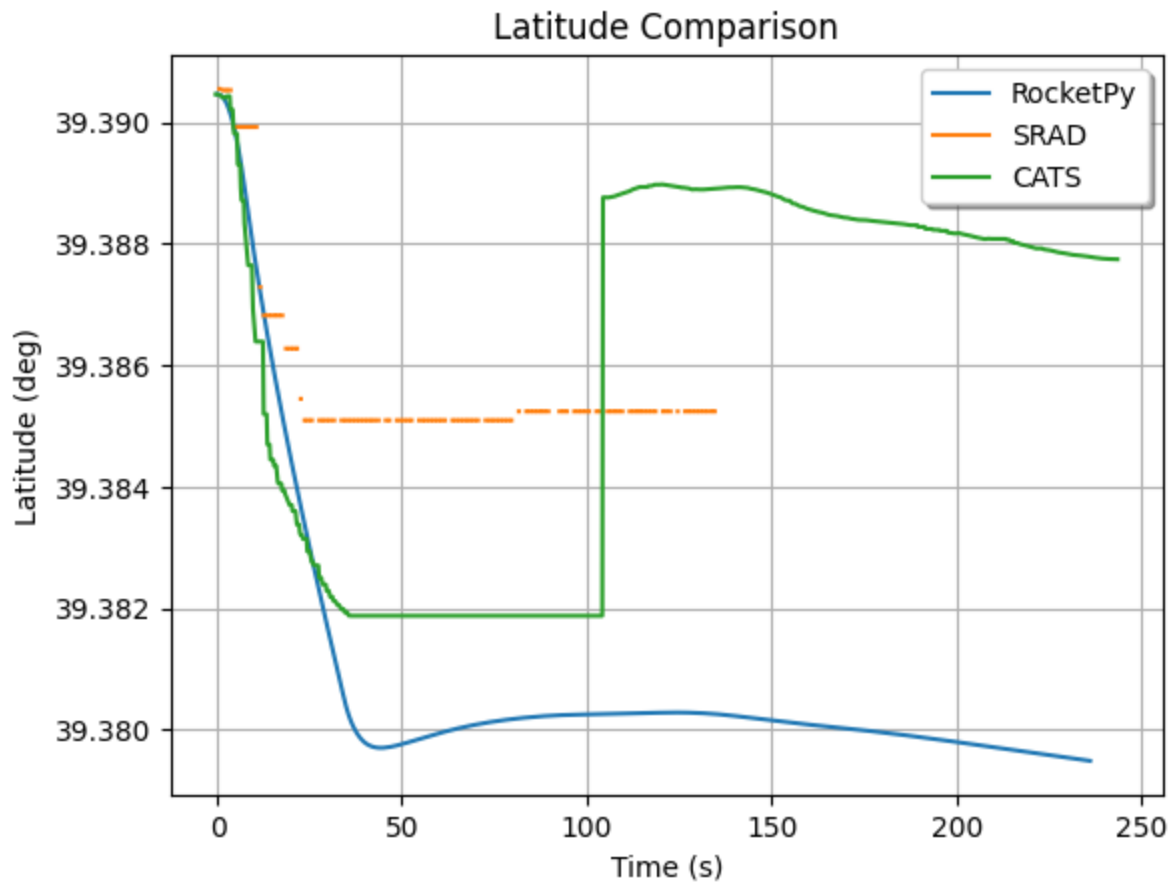
Actual apogee: 5279.38 m
 Simulated apogee: 5406.91 m
 Error: 127.53 m
 Percentage Error: 2.42%



GNSS

In [502...

```
Function.compare_plots(
    [
        (flight_reanalysis.latitude, "RocketPy"),
        (srad_flight.latitude, "SRAD"),
        (cats_vega_flight.latitude, "CATS")
    ],
    title="Latitude Comparison",
    xlabel="Time (s)",
    ylabel="Latitude (deg)",
)
Function.compare_plots(
    [
        (flight_reanalysis.longitude, "RocketPy"),
        (srad_flight.longitude, "SRAD"),
        (cats_vega_flight.longitude, "CATS")
    ],
    title="Longitude Comparison",
    xlabel="Time (s)",
    ylabel="Longitude (deg)",
)
```



Pressure

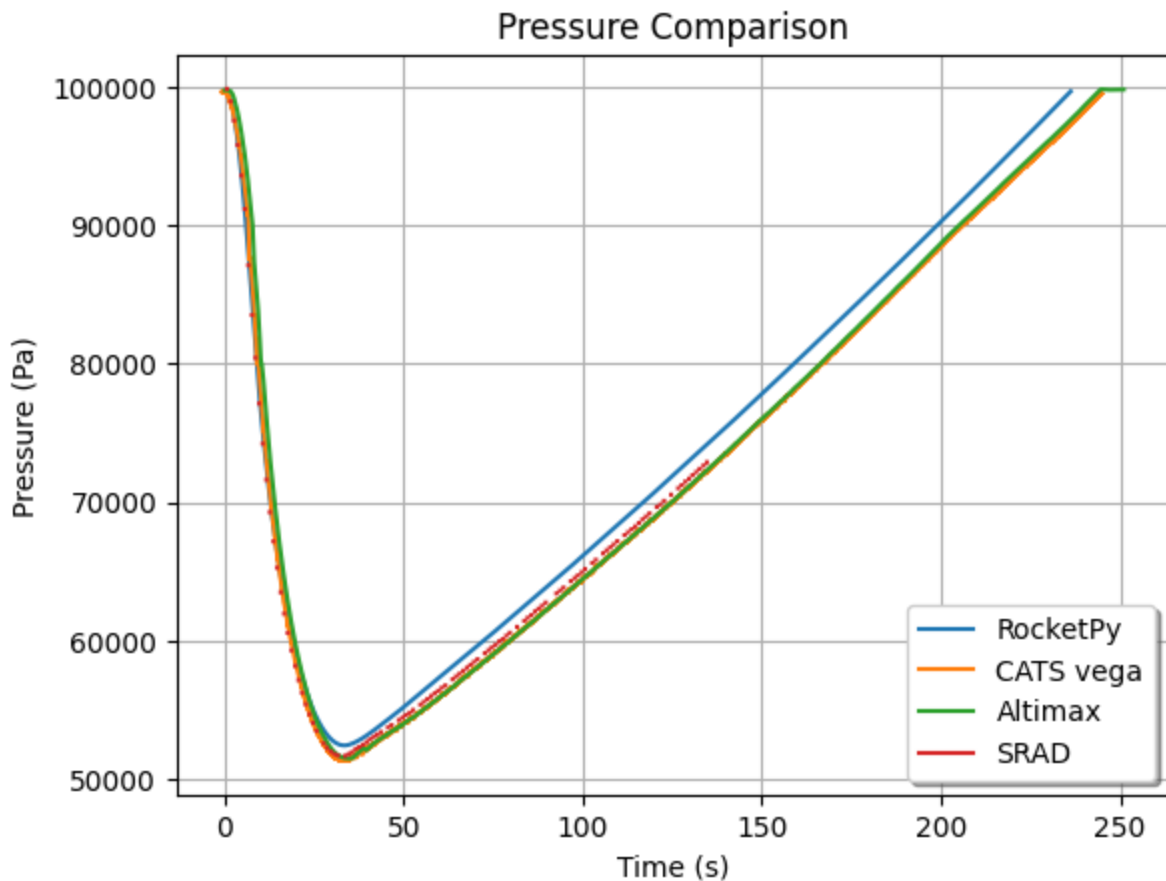
In [503...

```
pressure_actual = altimax_flight.pressure.min
pressure_simulated = flight_reanalysis.pressure.min
pressure_error = abs(pressure_actual - pressure_simulated)
pressure_percentage_error = pressure_error / pressure_actual * 100

print(f"Actual min pressure: {pressure_actual:.2f} Pa")
print(f"Simulated min pressure: {pressure_simulated:.2f} Pa")
print(f"Error: {pressure_error:.2f} Pa")
print(f"Percentage Error: {pressure_percentage_error:.2f}%")

Function.compare_plots(
    [
        (flight_reanalysis.pressure, "RocketPy"),
        (cats_vega_flight.pressure, "CATS vega"),
        (altimax_flight.pressure, "Altimax"),
        (srاد_flight.pressure*100, "SRAD")
    ],
    title="Pressure Comparison",
    xlabel="Time (s)",
    ylabel="Pressure (Pa)",
)
```

Actual min pressure: 51521.53 Pa
Simulated min pressure: 52452.64 Pa
Error: 931.11 Pa
Percentage Error: 1.81%



Acceleration

In [504...

```

acceleration_actual = altimax_flight.acceleration.crop([(0, 20)]).max
acceleration_simulated = flight_reanalysis.acceleration.crop([(0, 20)]).max
acceleration_error = abs(acceleration_actual - acceleration_simulated)
acceleration_percentage_error = acceleration_error / acceleration_actual * 100

print(f"Actual max acceleration during burn: {acceleration_actual:.2f} m/s^2")
print(f"Simulated max acceleration during burn: {acceleration_simulated:.2f} m/s^2")
print(f"Error: {acceleration_error:.2f} m/s^2")
print(f"Percentage Error: {acceleration_percentage_error:.2f}%")

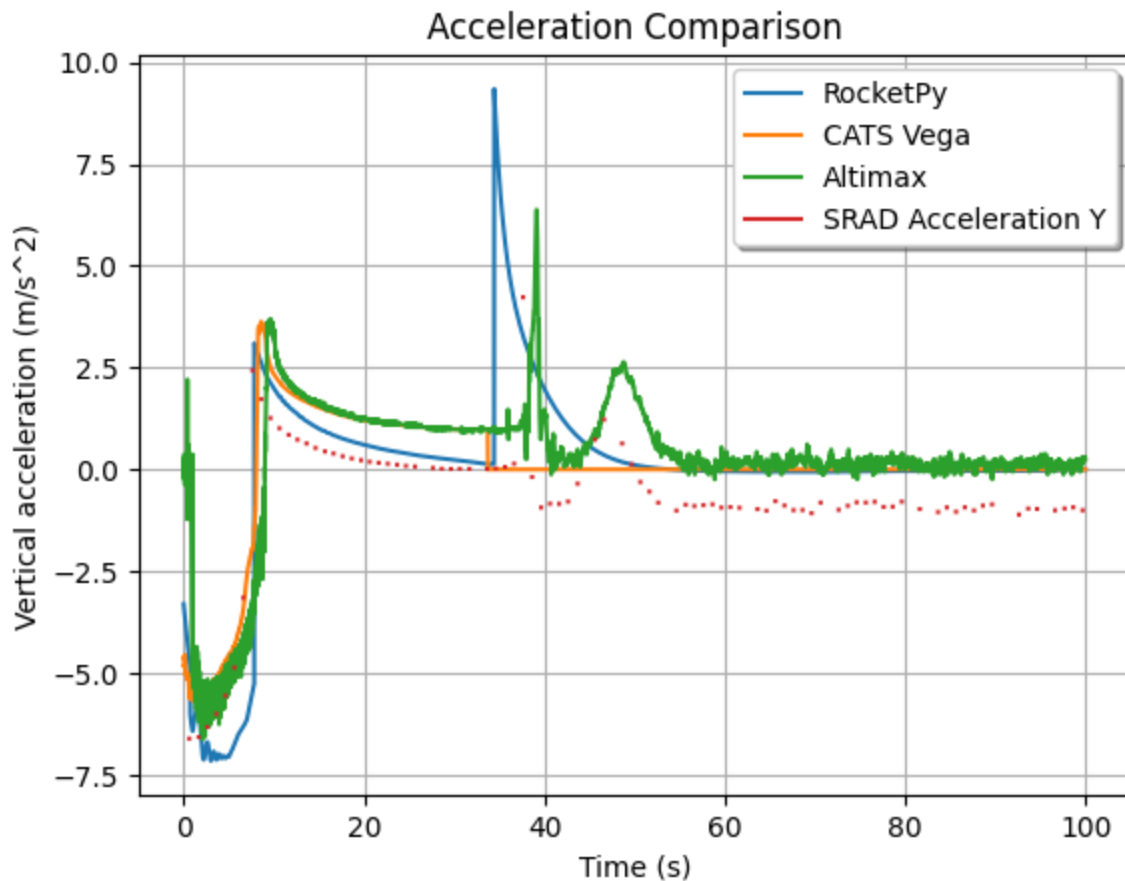
Function.compare_plots(
    [
        (flight_reanalysis.ay.crop([(0, 100)]), "RocketPy"),
        (cats_vega_flight.az.crop([(0, 100)]) / 10 * (-1), "CATS Vega"),
        (altimax_flight.acceleration.crop([(0, 100)]) / 10 * (-1), "Altimax"),
        (srad_flight.accel_y.crop([(0, 100)]), "SRAD Acceleration Y")
    ],
    title="Acceleration Comparison",
    xlabel="Time (s)",
    ylabel="Vertical acceleration (m/s^2)",
)

```

```

Actual max acceleration during burn: 66.22 m/s^2
Simulated max acceleration during burn: 55.42 m/s^2
Error: 10.80 m/s^2
Percentage Error: 16.30%

```



Speed

```
In [505... speed_actual = altimax_flight.speed.max
speed_simulated = flight_reanalysis.speed.max
speed_error = abs(speed_actual - speed_simulated)
speed_percentage_error = speed_error / speed_actual * 100

print(f"Actual max speed: {speed_actual:.2f} m/s2")
print(f"Simulated max speed: {speed_simulated:.2f} m/s2")
print(f"Error: {speed_error:.2f} m/s2")
print(f"Percentage Error: {speed_percentage_error:.2f}%")

Function.compare_plots(
    [
        (flight_reanalysis.vz.crop([(0, 100)]), "RocketPy"),
        (cats_vega_flight.speed.crop([(0, 100)]), "CATS Vega"),
        (altimax_flight.speed.crop([(0, 100)]), "Altimax")
    ],
    title="Speed Comparison",
    xlabel="Time (s)",
    ylabel="Speed (m/s)",
)
```

Actual max speed: 365.15 m/s2
 Simulated max speed: 365.66 m/s2
 Error: 0.51 m/s2
 Percentage Error: 0.14%

